THE HIDDEN CORE OF ROMANIAN: A GRAPH-THEORETIC ANALYSIS OF DICTIONARY DEFINITIONS

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Abstract

Dictionary definitions are inherently circular, as words are defined through other words. Consequently, a dictionary can only support lexical acquisition if part of its vocabulary is already known. Previous studies have shown that a dictionary represented as a directed graph contains a Kernel, from which all other words can be defined, and one or more minimal grounding sets (MinSets), the smallest sets of words that provide the lexical basis of the entire dictionary (Vincent-Lamarre et al., 2016).

In this study, we construct and analyse a graph representation of the *Dicționarul explicativ al limbii române* (DEX, second edition, digitised by dexonline), comprising 63,843 headwords and 600,473 definitional relations. Because Romanian exhibits extensive inflection and lexical ambiguity, automatic lemmatisation and headword assignment were required; these were performed using a large language model and evaluated against a stronger language model and two model-free baselines. Across all methods, the resulting graph structure proved highly stable.

Our analysis reveals that the DEX Kernel contains 21,216 words (33.2% of all headwords), while its MinSets comprise between 6,193 and 6,325 words (9.7–9.9%). These proportions are substantially larger than those obtained with the same procedure for dictionaries of eleven other languages (MinSets of 0.6–3.1%). We show that this difference is primarily attributable to reciprocal definitional relations, particularly prevalent in the lexicographic practice of the DEX. Despite its graph-theoretic derivation, the Kernel exhibits a close correspondence with independently established notions of fundamental vocabulary: it contains 98% of the Romanian equivalents of the Swadesh list, a standard list of about 200 basic meanings, and 94.5% of the fundamental lexical stock proposed by Graur (1954).

These findings suggest that the latent structure of dictionary definitions captures a linguistically meaningful lexical core. More broadly, the results show that graph-theoretic approaches can uncover fundamental vocabulary layers in morphologically rich languages.

Keywords: lexicography, dictionary graphs, symbol grounding, minimal grounding set, fundamental lexical stock, Romanian, DEX, large language models